ICS 253 – Discrete Structures
Chapter 1: Logic

Question bank: 90 multiple choice + 10 written problems

Questions from ICS 253 old quizzes and homework (AI_Work). **[new]** marks new questions written in the same style.

Part A: Multiple choice

Q1. [new]

Which of the following sentences is a proposition?

(A) Shut the door.

(B) Would you like some cake?

(C) Interest rates rose last year.

(D) What time is it?

Answer: (C). It is a declarative sentence that is either true or false; commands and questions have no truth value.

My answer

Q2. Quiz 1 Alrasasi

Let p: "This mobile is good" and q: "This mobile is expensive". The statement "This mobile is good but expensive" is written in logical symbols as:

(A) $p \vee q$

(B) $p \wedge q$

(C) $p \rightarrow q$

(D) $p \wedge \neg q$

Answer: (B). "But" expresses a conjunction: both p and q are true.

My answer

Q3. Quiz 1 (232) Sec 8

Let p: "I have a car", q: "I have a driving license", r: "I walk to school". The statement "I have a car and a driving license but I walk to school" is equivalent to:

(A) $(p \wedge q) \rightarrow r$

(B) $(p \wedge q) \rightarrow \neg r$

(C) $(p \wedge q) \wedge r$

(D) $(p \wedge q) \wedge \neg r$

Answer: (C). "and" and "but" are both conjunctions, so the statement is $(p \wedge q) \wedge r$.

My answer

Q4. [new]

Under which truth assignment do $p \vee q$ and $p \oplus q$ have different truth values?

(A) $p = \text{T}$, $q = \text{T}$

(B) $p = \text{T}$, $q = \text{F}$

(C) $p = \text{F}$, $q = \text{T}$

(D) $p = \text{F}$, $q = \text{F}$

Answer: (A). The inclusive or is true when both are true, while the exclusive or is false when both are true; in all other rows they agree.

My answer

Q5. [new]

Without parentheses, the compound proposition $\neg p \vee q \wedge r$ is evaluated as:

(A) $\neg(p \vee q) \wedge r$

(B) $(\neg p \vee q) \wedge r$

(C) $\neg p \vee (q \wedge r)$

(D) $\neg\big(p \vee (q \wedge r)\big)$

Answer: (C). Order of operations: $\neg$ first, then $\wedge$, then $\vee$.

My answer

Q6. Quiz 1 (232) Sec 3

How many rows appear in a truth table for the compound proposition $(p \wedge r \wedge s) \vee (q \wedge u) \vee (\neg w \wedge \neg u)$?

(A) 2^5

(B) 2^6

(C) 2^7

(D) 2^8

Answer: (B). The distinct variables are p, q, r, s, u, w: six variables give $2^6 = 64$ rows.

My answer

Q7. HW1 (261), Rosen 1.1 #32f

Consider the truth table of $(p \leftrightarrow q) \oplus (p \leftrightarrow \neg q)$. The final column is:

(A) T in every row (a tautology)

(B) F in every row (a contradiction)

(C) T only when p and q have the same truth value

(D) T only when p and q have different truth values

Answer: (A). $p \leftrightarrow \neg q$ always has the opposite value of $p \leftrightarrow q$, and the exclusive or of a proposition with its negation is always true.

My answer

Q8. Quiz 1 (232) Sec 3

The compound statement $u \rightarrow (p \rightarrow \neg w)$

(A) is a tautology.

(B) is False when $u \equiv$ F, $p \equiv$ T and $w \equiv$ F.

(C) is always False.

(D) is False only when $u \equiv$ T, $p \equiv$ T and $w \equiv$ T.

Answer: (D). The outer conditional is false only when $u =$ T and $p \rightarrow \neg w$ is false, i.e. $p =$ T and $\neg w =$ F ($w =$ T).

My answer

Q9. Quiz 1 Alrasasi

With p: "This mobile is good" and q: "This mobile is expensive", the statement "This mobile is good only if it is expensive" is:

(A) $q \rightarrow p$

(B) $p \rightarrow q$

(C) $p \leftrightarrow q$

(D) $\neg p \rightarrow \neg q$

Answer: (B). "p only if q" means $p \rightarrow q$.

My answer

Q10. **[new]**

Let p: "the applicant has parental permission" and c: "the applicant can enroll in the course". Which expression means "Having parental permission is a necessary condition for enrolling in the course"?

(A) $p \rightarrow c$

(B) $c \to p$

(C) $p \wedge c$

(D) $\neg c \to \neg p$

Answer: (B). "q is necessary for p" is $p \to q$; here enrolling requires permission, so $c \to p$. (D) is equivalent to $p \to c$.

My answer

Q11.

Quiz 1 Alrasasi

The inverse of the conditional statement "If you understand logic, then you will pass Math 210" is:

(A) If you pass Math 210, then you understand logic.

(B) If you do not pass Math 210, then you do not understand logic.

(C) If you do not understand logic, then you will not pass Math 210.

(D) If you understand logic, then you will not pass Math 210.

Answer: (C). The inverse of $p \to q$ is $\neg p \to \neg q$. (A) is the converse and (B) is the contrapositive.

My answer

Q12.

Quiz 1 (232) Sec 3

What is the contrapositive of the statement: "If n^2 is divisible by 3, then n is divisible by 3"?

(A) If n^2 is not divisible by 3, then n is not divisible by 3.

(B) If n is divisible by 3, then n^2 is divisible by 3.

(C) If n is not divisible by 3, then n^2 is not divisible by 3.

(D) If n is divisible by 3, then n^2 is not divisible by 3.

Answer: (C). The contrapositive of $p \to q$ is $\neg q \to \neg p$.

My answer

Q13.

Quiz 1 Alkhulaifi

The contrapositive and its truth value of the conditional statement "If 5 is a positive number, then $7 < 6$" are:

(A) If $7 \geq 6$, then 5 is not a positive number (False)

(B) If $7 \geq 6$, then 5 is a positive number (True)

(C) If 5 is not a positive number, then $7 \geq 6$ (True)

(D) If $7 < 6$, then 5 is not a positive number (False)

Answer: (A). The contrapositive is $\neg q \to \neg p$. Its hypothesis $7 \geq 6$ is true and its conclusion is false, so it is false (as is the original, T $\to$ F).

My answer

Q14. [new]

The variable p is true, q is false, and the truth value of r is unknown. Which expression has a truth value that is known?

(A) $p \wedge r$

(B) $q \to r$

(C) $p \to r$

(D) $r \vee q$

Answer: (B). The hypothesis q is false, so $q \to r$ is true whatever r is. The other three take the value of r.

My answer

Q15. Quiz 1 Alkhulaifi

The logical expression $(\neg p \vee q) \leftrightarrow (p \wedge \neg q)$ is

(A) a tautology

(B) equivalent to $p \vee q$

(C) neither a tautology nor a contradiction

(D) a contradiction

Answer: (D). $p \wedge \neg q \equiv \neg(\neg p \vee q)$, so the two sides always have opposite values and the biconditional is always false.

My answer

Q16. Quiz 1 Alkhulaifi

The truth assignment that shows the proposition $(r \wedge \neg q) \to (p \leftrightarrow q)$ is not a tautology is

(A) $p =$ T, $q = r =$ T

(B) $p =$ T, $q =$ F, $r =$ T

(C) $p =$ T, $q = r =$ F

(D) $p = q = r =$ F

Answer: (B). We need $r \wedge \neg q$ true ($r =$ T, $q =$ F) and $p \leftrightarrow q$ false ($p =$ T); this is the only falsifying row.

My answer

Q17. Quiz 1 (232) Sec 8

The statement $\big[(p \to q) \wedge (\neg p \to \neg q)\big]$ is equivalent to:

(A) $p \wedge q$

(B) $p \vee q$

(C) $p \oplus q$

(D) $p \leftrightarrow q$

Answer: (D). $\neg p \to \neg q \equiv q \to p$, and $(p \to q) \wedge (q \to p) \equiv p \leftrightarrow q$.

My answer

Q18. Quiz 1 Adel

The proposition $\neg\big(p \vee (\neg p \wedge q)\big)$ is logically equivalent to:

(A) $\neg p \wedge \neg q$

(B) $\neg p \vee \neg q$

(C) $\neg p \wedge q$

(D) $p \vee q$

Answer: (A). $p \vee (\neg p \wedge q) \equiv (p \vee \neg p) \wedge (p \vee q) \equiv p \vee q$, and $\neg(p \vee q) \equiv \neg p \wedge \neg q$ (the truth-table columns agree in all four rows).

My answer

Q19. HW1 (261), Rosen 1.3 #18

Which proposition is logically equivalent to $p \to q$?

(A) $q \to p$

(B) $\neg p \to \neg q$

(C) $\neg q \to \neg p$

(D) $p \wedge \neg q$

Answer: (C). A conditional is equivalent to its contrapositive; (A) converse and (B) inverse are not, and (D) is its negation.

My answer

Q20. HW1 (261), Rosen 1.3 #8a

Using De Morgan's laws, the negation of "Kwame will take a job in industry or go to graduate school" is:

(A) Kwame will not take a job in industry or will not go to graduate school.

(B) Kwame will not take a job in industry and will not go to graduate school.

(C) Kwame will take a job in industry and will not go to graduate school.

(D) If Kwame does not take a job in industry, then he will go to graduate school.

Answer: (B). $\neg(p \vee q) \equiv \neg p \wedge \neg q$.

My answer

Q21. Quiz 1 (232) Sec 8

Which law, applied once, proves that $\mathrm{F} \wedge \mathrm{T} \equiv \mathrm{F}$?

(A) Identity law

(B) Idempotent law

(C) Absorption law

(D) De Morgan's law

Answer: (A). Identity law $p \wedge \mathrm{T} \equiv p$ with $p = \mathrm{F}$. (The original was "select all": after commuting, the domination and negation/complement laws also give it.)

My answer

Q22. Quiz 1 Alkhulaifi

Using the laws of propositional logic, the proposition $(p \wedge q \wedge \neg r) \vee (p \wedge \neg q \wedge \neg r)$ is equivalent to

(A) $\neg r$

(B) $p \wedge \neg r$

(C) $p \vee \neg r$

(D) $p \wedge r$

Answer: (B). Factor out $p \wedge \neg r$ (commutative, distributive): $(p \wedge \neg r) \wedge (q \vee \neg q) \equiv (p \wedge \neg r) \wedge \mathrm{T} \equiv p \wedge \neg r$.

My answer

Q23. Quiz 1 Adel

Using the laws of propositional logic, the proposition $\neg p \to (p \to q)$ is

(A) a tautology

(B) a contradiction

(C) a contingency (neither)

(D) equivalent to q

Answer: (A). $\neg p \to (\neg p \vee q) \equiv \neg\neg p \vee (\neg p \vee q) \equiv (p \vee \neg p) \vee q \equiv \mathrm{T} \vee q \equiv \mathrm{T}$.

My answer

Q24.

HW1 (232), Rosen 1.3 #12a

To show that $\big[\neg p \wedge (p \vee q)\big] \to q$ is a tautology, the hypothesis is first simplified with the distributive, complement and identity laws. The hypothesis $\neg p \wedge (p \vee q)$ simplifies to:

(A) $\neg p \wedge q$

(B) q

(C) $\neg p$

(D) $p \vee q$

Answer: (A). $(\neg p \wedge p) \vee (\neg p \wedge q) \equiv \mathrm{F} \vee (\neg p \wedge q) \equiv \neg p \wedge q$. Then $(\neg p \wedge q) \to q \equiv p \vee \neg q \vee q \equiv \mathrm{T}$.

My answer

Q25.

Quiz 1 (232) Sec 3

The following steps, in the correct order, prove $(r \wedge \neg s) \vee s \equiv \neg r \to s$:

[1] $\equiv s \vee (r \wedge \neg s)$	Commutative
[2] $\equiv r \vee s$	Commutative
[3] $\equiv (s \vee r) \wedge \mathrm{T}$	Complement (OR tautology)
[4] $\equiv \neg r \to s$	Conditional identity
[5] $(r \wedge \neg s) \vee s$	Given
[6] $\equiv s \vee r$	Identity
[7] $\equiv (s \vee r) \wedge (s \vee \neg s)$	Distributive
[8] $\equiv \neg\neg r \vee s$	Double negation

The proof starts with [5] and ends with [4]. The correct order of the six middle steps is:

(A) 1, 7, 3, 6, 2, 8

(B) 7, 1, 3, 6, 8, 2

(C) 1, 3, 7, 6, 2, 8

(D) 1, 7, 6, 3, 2, 8

Answer: (A). Sequence 5, 1, 7, 3, 6, 2, 8, 4: commute, distribute, $s \vee \neg s \equiv \mathrm{T}$, drop $\wedge\,\mathrm{T}$, commute, $r \equiv \neg\neg r$, conditional identity.

My answer

Q26.

[new]

The domain is the set of positive integers. Which of the following is a proposition?

(A) x is odd.

(B) $x + 1 > x$

(C) $\forall x\,(x + 1 > x)$

(D) $x^2 \geq 1$

Answer: (C). A statement with a free variable is a predicate even when it holds for every value, as in (B) and (D). Binding x with a quantifier gives a proposition (a true one).

My answer

Q27. HW1 (261), Rosen 1.4 #13

The domain is the set of all integers. Which of the following statements are true?

(a) $\forall n\,(n + 1 > n)$ (b) $\exists n\,(2n = 3n)$ (c) $\exists n\,(n = -n)$ (d) $\forall n\,(3n \leq 4n)$

(A) All four

(B) (a), (b) and (c) only

(C) (a) and (c) only

(D) (a) and (d) only

Answer: (B). (b) and (c) hold for $n = 0$. (d) is false: $n = -1$ gives $-3 \leq -4$, which is false.

My answer

Q28. HW1 (232), Rosen 1.4 #20c

The domain of $P(x)$ is $\{-5, -3, -1, 1, 3, 5\}$. Without quantifiers, $\forall x\,\big((x \neq 1) \to P(x)\big)$ is equivalent to:

(A) $P(-5) \wedge P(-3) \wedge P(-1) \wedge P(3) \wedge P(5)$

(B) $P(-5) \vee P(-3) \vee P(-1) \vee P(3) \vee P(5)$

(C) $P(-5) \wedge P(-3) \wedge P(-1) \wedge \neg P(1) \wedge P(3) \wedge P(5)$

(D) $\neg P(1)$

Answer: (A). For $x = 1$ the conditional has a false hypothesis and is true; for every other x it reduces to $P(x)$.

My answer

Q29. HW1 (232), Rosen 1.4 #20d

The domain of $P(x)$ is $\{-5, -3, -1, 1, 3, 5\}$. Without quantifiers, $\exists x\,\big((x \geq 0) \wedge P(x)\big)$ is equivalent to:

(A) $P(1) \wedge P(3) \wedge P(5)$

(B) $P(1) \vee P(3) \vee P(5)$

(C) $P(-5) \vee P(-3) \vee P(-1)$

(D) $P(0) \lor P(1) \lor P(3) \lor P(5)$

Answer: (B). Only $1, 3, 5$ satisfy $x \geq 0$, and "there exists" becomes a disjunction (0 is not in the domain).

My answer

Q30.

HW1 (261), Rosen 1.4 #19c

The domain of $P(x)$ is $\{1, 2, 3, 4, 5\}$. Using only negations, disjunctions and conjunctions, $\neg\exists x\, P(x)$ is:

(A) $\neg P(1) \lor \neg P(2) \lor \neg P(3) \lor \neg P(4) \lor \neg P(5)$

(B) $\neg P(1) \land \neg P(2) \land \neg P(3) \land \neg P(4) \land \neg P(5)$

(C) $\neg\big(P(1) \land P(2) \land P(3) \land P(4) \land P(5)\big)$

(D) $P(1) \land P(2) \land P(3) \land P(4) \land P(5)$

Answer: (B). $\neg(P(1) \lor \cdots \lor P(5)) \equiv \neg P(1) \land \cdots \land \neg P(5)$ by De Morgan's law. (C) is equivalent to (A).

My answer

Q31.

HW1 (261), Rosen 1.4 #19e

The domain of $P(x)$ is $\{1, 2, 3, 4, 5\}$. Without quantifiers, $\forall x\,\big((x \neq 3) \to P(x)\big) \lor \exists x\, \neg P(x)$ is:

(A) $\big(P(1) \land P(2) \land P(4) \land P(5)\big) \lor \big(\neg P(1) \lor \neg P(2) \lor \neg P(3) \lor \neg P(4) \lor \neg P(5)\big)$

(B) $\big(P(1) \land P(2) \land P(4) \land P(5)\big) \lor \big(\neg P(1) \land \neg P(2) \land \neg P(3) \land \neg P(4) \land \neg P(5)\big)$

(C) $\big(P(1) \lor P(2) \lor P(4) \lor P(5)\big) \lor \big(\neg P(1) \lor \neg P(2) \lor \neg P(3) \lor \neg P(4) \lor \neg P(5)\big)$

(D) $\big(P(1) \land P(2) \land P(3) \land P(4) \land P(5)\big) \lor \big(\neg P(1) \lor \neg P(2) \lor \neg P(3) \lor \neg P(4) \lor \neg P(5)\big)$

Answer: (A). The universal part skips $x = 3$; the existential part is a disjunction over the whole domain. (In fact the expression is always true: if one of $P(1), P(2), P(4), P(5)$ is false, the second part is true.)

My answer

Q32.

Quiz 1 (232) Sec 8

The domain is the set of integers $\mathbb{Z}$. Determine whether $\exists x\,\big[(x < 0) \to (x = x^2)\big]$ is true or false, with justification.

(A) False, for $x = 0$

(B) False, for $x = -1$

(C) True, for $x = 0$

(D) True, for $x = -1$

Answer: (C). For $x = 0$ the hypothesis $0 < 0$ is false, so the conditional is true; one example proves an existential statement. ($x = -1$ makes it false, but that does not decide an existential statement.)

My answer

Q33. [new]

Which of the following logical expressions is a proposition (has no free variable)?

(A) $\forall x\, P(x) \land Q(x)$

(B) $\exists x\, \big(P(x) \land Q(x)\big)$

(C) $P(x) \to \forall y\, Q(y)$

(D) $\exists x\, P(x) \lor Q(y)$

Answer: (B). In (A) the scope of $\forall x$ is only $P(x)$, so the x in $Q(x)$ is free; (C) and (D) also have free variables.

My answer

Q34. Quiz 1 Alrasasi

The domain is the set of students at a university, $E(x)$: "x is enrolled in Math 101 class", $T(x)$: "x took the test". The statement "Every student enrolled in Math 101 class did not take the test" is:

(A) $\forall x\, \big(E(x) \land \neg T(x)\big)$

(B) $\forall x\, \big(E(x) \to \neg T(x)\big)$

(C) $\neg\forall x\, \big(E(x) \to T(x)\big)$

(D) $\exists x\, \big(E(x) \land \neg T(x)\big)$

Answer: (B). "Every E is not T" uses a universal quantifier with a conditional. (A) claims that every student is enrolled.

My answer

Q35. HW1 (261), Rosen 1.4 #9(a,c)

$P(x)$: "x can speak Russian", $Q(x)$: "x knows C++"; the domain is all students at your school. Translate (a) "There is a student at your school who can speak Russian and who knows C++" and (c) "Every student at your school either can speak Russian or knows C++".

(A) (a) $\exists x\, (P(x) \land Q(x))$; (c) $\forall x\, (P(x) \lor Q(x))$

(B) (a) $\exists x\, (P(x) \to Q(x))$; (c) $\forall x\, (P(x) \lor Q(x))$

(C) (a) $\exists x\, (P(x) \land Q(x))$; (c) $\forall x\, (P(x) \to Q(x))$

(D) (a) $\forall x\, (P(x) \land Q(x))$; (c) $\exists x\, (P(x) \lor Q(x))$

Answer: (A). "There is … and" is $\exists$ with $\land$; "every … or" is $\forall$ with $\lor$.

My answer

Q36. Quiz 1 (232) Sec 3

Let $B(f)$ be "Fax f is busy", $L(j)$ be "Fax job j is lost", and $Q(j)$ be "Fax job j is queued". Then $\big(\forall f\, B(f) \wedge \forall j\, Q(j)\big) \to \exists j\, L(j)$ is translated to English as:

(A) If every fax is busy and every job is queued, then all jobs are lost.

(B) If every fax is busy and every job is queued, then no job is lost.

(C) If some fax is busy and some job is queued, then some job is lost.

(D) If every fax is busy and every job is queued, then some job is lost.

Answer: (D). Both hypotheses are universal and the conclusion is existential.

My answer

Q37. HW1 (232), Rosen 1.4 #26

The domain is all people. Let $S(x)$: "x is in your school" and $V(x, y)$: "x has visited y". The statement "Someone in your school has visited Uzbekistan" is:

(A) $\exists x \big(S(x) \to V(x, \text{Uzbekistan})\big)$

(B) $\exists x \big(S(x) \wedge V(x, \text{Uzbekistan})\big)$

(C) $\forall x \big(S(x) \to V(x, \text{Uzbekistan})\big)$

(D) $\forall x \big(S(x) \wedge V(x, \text{Uzbekistan})\big)$

Answer: (B). An existential statement restricted to people in the school uses $\wedge$. (A) is true as soon as one person is not in the school.

My answer

Q38. Quiz 1 Alrasasi

What is the negation of the statement "All male students are smart"?

(A) All male students are not smart.

(B) No male student is smart.

(C) There exists a male student who is not smart.

(D) There exists a male student who is smart.

Answer: (C). $\neg\forall x\, S(x) \equiv \exists x\, \neg S(x)$ (domain: male students).

My answer

Q39. Quiz 1 Adel

The domain is the set of students who showed up for an exam, $P(x)$: "x showed up with a pencil", $C(x)$: "x showed up with a calculator". The negation of "Every student who showed up with a calculator also had a pencil", with no negation outside a quantifier or a connective, is:

(A) $\forall x \left(C(x) \to \neg P(x)\right)$

(B) $\exists x \left(C(x) \wedge \neg P(x)\right)$

(C) $\exists x \left(\neg C(x) \wedge P(x)\right)$

(D) $\forall x \left(\neg C(x) \to \neg P(x)\right)$

Answer: (B). $\neg\forall x(C(x) \to P(x)) \equiv \exists x\, \neg(\neg C(x) \vee P(x)) \equiv \exists x(C(x) \wedge \neg P(x))$: some student had a calculator but no pencil.

My answer

Q40. HW1 (261), Rosen 1.5 #3(a,e)

$Q(x, y)$: "x has sent an e-mail message to y"; the domain is all students in your class. Express (a) $\exists x \exists y\, Q(x, y)$ and (e) $\forall y \exists x\, Q(x, y)$ in English.

(A) (a) Some student has sent an e-mail to some student; (e) every student has received an e-mail from some student.

(B) (a) Some student has sent an e-mail to some student; (e) some student has sent an e-mail to every student.

(C) (a) Every student has sent an e-mail to some student; (e) every student has received an e-mail from some student.

(D) (a) Some student has sent an e-mail to every student; (e) every student has sent an e-mail to some student.

Answer: (A). In (e) the outer variable y is the receiver: for every receiver there is a sender.

My answer

Q41. HW1 (261), Rosen 1.5 #5d

$W(x, y)$: "student x has visited website y". The statement $\exists y \left(W(\text{Ashok Puri}, y) \wedge W(\text{Cindy Yoon}, y)\right)$ means:

(A) Ashok Puri and Cindy Yoon have visited every website.

(B) There is a website that both Ashok Puri and Cindy Yoon have visited.

(C) Ashok Puri has visited some website and Cindy Yoon has visited some (possibly different) website.

(D) There is a website that Ashok Puri or Cindy Yoon has visited.

Answer: (B). The same y appears in both conjuncts, so it is one common website.

My answer

Q42. HW1 (261), Rosen 1.5 #5e

$W(x, y)$: "student x has visited website y". The statement

$$\exists y \, \forall z \, \big(y \neq \text{David Belcher} \wedge (W(\text{David Belcher}, z) \rightarrow W(y, z))\big)$$

means:

(A) David Belcher has visited every website that some other student has visited.

(B) There is a student other than David Belcher who has visited every website that David Belcher has visited.

(C) Every student other than David Belcher has visited some website that David Belcher has visited.

(D) There is a website visited by every student other than David Belcher.

Answer: (B). y is a student different from David, and every website z David visited was also visited by y.

My answer

Q43. HW1 (261), Rosen 1.5 #16a

A class has 1 math freshman, 12 math sophomores, 15 CS sophomores, 2 math juniors, 2 CS juniors and 1 CS senior. With domain the students in the class and $J(x)$: "x is a junior", the statement "There is a student in the class who is a junior" and its truth value are:

(A) $\forall x \, J(x)$; false

(B) $\exists x \, J(x)$; true

(C) $\exists x \, J(x)$; false

(D) $\forall x \, J(x)$; true

Answer: (B). There are four juniors in the class.

My answer

Q44. HW1 (261), Rosen 1.5 #16e

Same class as before (math: 1 freshman, 12 sophomores, 2 juniors; CS: 15 sophomores, 2 juniors, 1 senior). Let $Q(s, y, m)$: "student s is in year y and has major m". "There is a major such that there is a student in the class in every year of study with that major" is expressed and evaluated as:

(A) $\exists m \, \forall y \, \exists s \, Q(s, y, m)$; true

(B) $\exists m \, \forall y \, \exists s \, Q(s, y, m)$; false

(C) $\forall m \, \exists y \, \exists s \, Q(s, y, m)$; false

(D) $\exists s \, \forall y \, \exists m \, Q(s, y, m)$; true

Answer: (B). Math has no senior and CS has no freshman, so no major covers all four years.

My answer

Q45. HW1 (261), Rosen 1.5 #26(a,i,h)

Let $Q(x, y)$ mean "$x + y = x - y$", both variables ranging over all integers. The truth values of (a) $Q(1, 1)$, (i) $\forall x \forall y\, Q(x, y)$ and (h) $\forall y \exists x\, Q(x, y)$ are, in order:

(A) F, F, F

(B) F, F, T

(C) T, F, T

(D) F, T, F

Answer: (A). $x + y = x - y \iff y = 0$. (a) $2 \neq 0$; (i) fails at $y = 1$; (h) fails for $y = 1$, since no x works.

My answer

Q46. HW1 (261), Rosen 1.5 #40

The domain of all variables is the set of integers. Which of the following universally quantified statements have a counterexample?

(a) $\forall x \exists y\, (x = 1/y)$ (b) $\forall x \exists y\, (y^2 - x < 100)$ (c) $\forall x \forall y\, (x^2 \neq y^3)$

(A) (a) and (c) only

(B) All three: e.g. $x = 2$; $x = -100$; $x = y = 1$

(C) (c) only

(D) (a) and (b) only

Answer: (B). (a) $1/y = 2$ has no integer solution; (b) with $x = -100$ we need $y^2 < 0$; (c) $1^2 = 1^3$.

My answer

Q47. Quiz 2 (232) Sec 3

The domain of all variables is the set of positive rational numbers $\mathbb{Q}^+$. Consider

(i) $\neg \forall a \exists b \left(a = \frac{1}{b}\right)$ (ii) $\forall a \exists b\, (b^2 - a < 100)$ (iii) $\forall a \forall b\, (a^2 \neq b^3)$.

Then,

(A) (i) is True, (ii) is False, (iii) is False.

(B) (i) is True, (ii) is False, (iii) is True.

(C) (i) is False, (ii) is True, (iii) is False.

(D) (i) is True, (ii) is True, (iii) is True.

Answer: (C). (i) $\forall a \exists b$ holds with $b = 1/a$, so its negation is false; (ii) choose b with $b^2 < 100$; (iii) fails at $a = b = 1$.

My answer

Q48.

Quiz 2 (232) Sec 3

Let $Q(x, y)$ denote "$x + y = 0$", the domain being all real numbers. The truth values of (i) $\exists y \forall x\, Q(x, y)$ and (ii) $\forall x \exists y\, Q(x, y)$ are:

(A) (i) True, (ii) False

(B) (i) True, (ii) True

(C) (i) False, (ii) False

(D) (i) False, (ii) True

Answer: (D). (ii) take $y = -x$. (i) no single y works for all x: $x = 1 - y$ gives $x + y = 1$.

My answer

Q49.

Quiz 1 Alrasasi

The domain is the set of real numbers. Is $\forall x \exists y\, (xy > 1)$ true or false?

(A) True: take $y = 2/x$.

(B) False: take $x = 0$.

(C) True: take $y = x + 1$.

(D) False: take $x = 1$.

Answer: (B). For $x = 0$, $0 \cdot y = 0 > 1$ fails for every y.

My answer

Q50.

Quiz 1 Alrasasi

The domain is the set of real numbers. Is $\exists x \exists y\, \big((x + y = y) \wedge (x \neq 0)\big)$ true or false?

(A) True: take $x = 1$, $y = 0$.

(B) True: take $x = 0$, $y = 1$.

(C) False: $x + y = y$ forces $x = 0$, contradicting $x \neq 0$.

(D) False: it holds only when $y = 0$.

Answer: (C). $(x = 0) \wedge (x \neq 0)$ is false for every real x.

My answer

Q51.

Quiz 1 Alkhulaifi

If the domain is the set of all integers, which of the following statements is FALSE?

(I) $\forall n \exists m\, (n^2 < m)$ (II) $\exists n \forall m\, (n < m^2)$ (III) $\exists n \exists m\, (n + m = 4 \wedge n - m = 1)$

(A) II and III only

(B) All of them

(C) I only

(D) III only

Answer: (D). (I) $m = n^2 + 1$; (II) $n = -1$; (III) the system gives $n = 5/2$, which is not an integer.

My answer

Q52.

Quiz 2 (232) Sec 8

Let the domain be all integers. Then $\exists x \exists y \left[(x < y) \to (x + 2 > y + 1)\right]$ is

(A) False, for $x = 1$, $y = 2$

(B) False, for $x = 2$, $y = 1$

(C) True, for $x = 1$, $y = 2$

(D) True, for $x = 2$, $y = 1$

Answer: (D). For $x = 2$, $y = 1$ the hypothesis $2 < 1$ is false, so the conditional is (vacuously) true. For $x = 1$, $y = 2$ it is T $\to$ F.

My answer

Q53.

Quiz 2 (232) Sec 3

The translation of $\forall x \forall y \left((x > 0) \wedge (y < 0) \to (xy < 0)\right)$, where the domain for both variables is all real numbers, is:

(A) The product of any two real numbers is either a negative or a positive real number.

(B) The product of any two negative real numbers is always a positive real number.

(C) The product of any two positive real numbers is always a positive real number.

(D) The product of a positive real number and a negative real number is always a negative real number.

Answer: (D). The hypothesis says x is positive and y is negative; the conclusion says xy is negative.

My answer

Q54.

Quiz 2 (232) Sec 3

$F(x)$: "x is female", $P(x)$: "x is a parent", $M(x, y)$: "x is the mother of y". Which statement means "If a person is female and is a parent, then this person is someone's mother"?

(A) $\exists x \forall y \left((F(x) \wedge P(x)) \to M(x, y)\right)$

(B) $\forall x \exists y \left((F(x) \wedge P(x)) \wedge M(x, y)\right)$

(C) $\exists x \forall y \left((F(x) \wedge P(x)) \wedge M(x, y)\right)$

(D) $\forall x \exists y \left((F(x) \wedge P(x)) \to M(x, y)\right)$

Answer: (D). For every person x: if x is a female parent, then there is some y whose mother is x.

My answer

Q55.

Quiz 2 (232) Sec 3

For any two predicates $P(w, f)$ and $Q(f, a)$, the negation of $\forall w \exists a \forall f \left(\neg P(w, f) \vee \neg Q(f, a)\right)$ is:

(A) $\exists w \forall a \exists f \left(\neg P(w, f) \vee \neg Q(f, a)\right)$

(B) $\exists w \forall a \exists f \left(P(w, f) \wedge Q(f, a)\right)$

(C) $\exists w \forall a \exists f \left(\neg P(w, f) \wedge \neg Q(f, a)\right)$

(D) $\exists w \forall a \exists f \left(P(w, f) \vee Q(f, a)\right)$

Answer: (B). Each quantifier flips, and $\neg(\neg P \vee \neg Q) \equiv P \wedge Q$.

My answer

Q56.

Quiz 2 (232) Sec 8

Let $P(x, y)$ be "x knows y" and $C(x)$ be "x is in our class", the domain being all KFUPM students. The statement "Nobody knows everyone in our class" can be written as:

(A) $\forall x \exists y \left[C(y) \wedge \neg P(x, y)\right]$

(B) $\forall x \exists y \left[C(y) \rightarrow \neg P(x, y)\right]$

(C) $\exists x \forall y \left[C(y) \rightarrow \neg P(x, y)\right]$

(D) $\neg \exists x \forall y \left[C(y) \wedge P(x, y)\right]$

Answer: (A). $\neg \exists x \forall y \left(C(y) \rightarrow P(x, y)\right) \equiv \forall x \exists y \neg(\neg C(y) \vee P(x, y)) \equiv \forall x \exists y \left(C(y) \wedge \neg P(x, y)\right)$.

My answer

Q57.

HW2 (232), Rosen 1.5 #10d

Let $F(x, y)$ be "x can fool y", the domain being all people. "There is no one who can fool everybody" is:

(A) $\neg \forall x \forall y \, F(x, y)$

(B) $\forall x \exists y \, \neg F(x, y)$

(C) $\exists x \forall y \, \neg F(x, y)$

(D) $\neg \exists x \exists y \, F(x, y)$

Answer: (B). $\neg \exists x \forall y \, F(x, y) \equiv \forall x \exists y \, \neg F(x, y)$. (The homework answer was (A), which only says that not everyone can fool everybody.)

My answer

Q58. HW2 (232), Rosen 1.5 #10e

With $F(x, y)$: "x can fool y", "Everyone can be fooled by somebody" is:

(A) $\forall x \exists y\, F(x, y)$

(B) $\forall y \exists x\, F(x, y)$

(C) $\exists x \forall y\, F(x, y)$

(D) $\exists y \forall x\, F(x, y)$

Answer: (B). For every victim y there is a fooler x. (A) says everyone can fool somebody.

My answer

Q59. HW2 (232), Rosen 1.5 #10f

With $F(x, y)$: "x can fool y", "No one can fool both Fred and Jerry" is:

(A) $\neg\exists x \left(F(x, \text{Fred}) \wedge F(x, \text{Jerry})\right)$

(B) $\exists x \left(\neg F(x, \text{Fred}) \wedge \neg F(x, \text{Jerry})\right)$

(C) $\forall x \left(\neg F(x, \text{Fred}) \wedge \neg F(x, \text{Jerry})\right)$

(D) $\neg\exists x \left(F(x, \text{Fred}) \vee F(x, \text{Jerry})\right)$

Answer: (A). Equivalently $\forall x(\neg F(x, \text{Fred}) \vee \neg F(x, \text{Jerry}))$. (C) and (D) say no one can fool either of them.

My answer

Q60. HW2 (232), Rosen 1.5 #10g

With $F(x, y)$: "x can fool y", "Nancy can fool exactly two people" is:

(A) $F(\text{Nancy}, A) \wedge F(\text{Nancy}, B)$

(B) $\exists y \exists z \left(F(\text{Nancy}, y) \wedge F(\text{Nancy}, z)\right)$

(C) $\exists y \exists z \left(y \neq z \wedge F(\text{Nancy}, y) \wedge F(\text{Nancy}, z)\right)$

(D) $\exists y \exists z \left(y \neq z \wedge F(\text{Nancy}, y) \wedge F(\text{Nancy}, z) \wedge \forall w \left(F(\text{Nancy}, w) \rightarrow (w = y \vee w = z)\right)\right)$

Answer: (D). We need two different people and nobody else. (C) means "at least two". (The homework answer was (A).)

My answer

Q61. HW2 (232), Rosen 1.5 #28(a,c,f,h)

The domain of each variable is all real numbers. Which of these statements are true?

(a) $\forall x \exists y\, (x^2 = y)$ (c) $\exists x \forall y\, (xy = 0)$ (f) $\exists x \forall y\, (y \neq 0 \rightarrow xy = 1)$ (h) $\exists x \exists y\, (x{+}2y = 2 \wedge 2x{+}4y = 5)$

(A) (a) and (c) only

(B) (a), (c) and (f) only

(C) All four

(D) (a) only

Answer: (A). (a) $y = x^2$; (c) $x = 0$; (f) is false because one x would need $x = 1/y$ for every y; (h) $2(x + 2y) = 4 \neq 5$. (The homework said (f) is true with $x = 1/y$, but x is chosen before y.)

My answer

Q62.

HW2 (232), Rosen 1.5 #32c

The negation of $\exists x \exists y \left(Q(x, y) \leftrightarrow Q(y, x)\right)$, with every negation immediately before a predicate, is:

(A) $\forall x \forall y \left(Q(x, y) \wedge \neg Q(y, x)\right)$

(B) $\forall x \forall y \left(Q(x, y) \leftrightarrow \neg Q(y, x)\right)$

(C) $\exists x \exists y \left(\neg Q(x, y) \leftrightarrow \neg Q(y, x)\right)$

(D) $\forall x \forall y \left(\neg Q(x, y) \leftrightarrow \neg Q(y, x)\right)$

Answer: (B). The quantifiers flip and $\neg(a \leftrightarrow b) \equiv a \leftrightarrow \neg b$. (The homework answer stopped at the form $\forall x \forall y \, \neg(Q(x, y) \leftrightarrow Q(y, x))$.)

My answer

Q63.

HW2 (232), Rosen 1.5 #32d

The negation of $\forall y \exists x \exists z \left(T(x, y, z) \vee Q(x, y)\right)$, with every negation immediately before a predicate, is:

(A) $\exists y \forall x \forall z \left(\neg T(x, y, z) \wedge \neg Q(x, y)\right)$

(B) $\exists y \forall x \forall z \left(\neg T(x, y, z) \vee \neg Q(x, y)\right)$

(C) $\forall y \exists x \exists z \left(\neg T(x, y, z) \wedge \neg Q(x, y)\right)$

(D) $\exists y \exists x \exists z \left(\neg T(x, y, z) \wedge \neg Q(x, y)\right)$

Answer: (A). Each quantifier changes type and De Morgan turns $\neg(T \vee Q)$ into $\neg T \wedge \neg Q$.

My answer

Q64.

HW2 (232), Rosen 1.5 #34

For which common domain is $\forall x \forall y \left((x \neq y) \rightarrow \forall z \left((z = x) \vee (z = y)\right)\right)$ true, and for which is it false?

(A) True for $\{0, 1\}$; false for $\{0, 1, 2\}$

(B) True for $\{0, 1, 2\}$; false for $\{0, 1\}$

(C) True for every nonempty domain

(D) False for every domain

Answer: (A). The statement says the domain has at most two elements. In $\{0, 1, 2\}$ take $x = 0$, $y = 1$, $z = 2$.

My answer

Q65.

Quiz 2 Alrasasi

What must we prove in order to show that the following statement is false: "There are two distinct integers x and y such that $\frac{1}{x} + \frac{1}{y}$ is an integer"?

(A) For all integers x and y, if $x \neq y$, then $\frac{1}{x} + \frac{1}{y}$ is not an integer.

(B) For all integers x and y, $x \neq y$ and $\frac{1}{x} + \frac{1}{y}$ is not an integer.

(C) There are two distinct integers x and y such that $\frac{1}{x} + \frac{1}{y}$ is not an integer.

(D) For all integers x and y, if $\frac{1}{x} + \frac{1}{y}$ is not an integer, then $x \neq y$.

Answer: (A). The statement is $\exists x \exists y\,(x \neq y \wedge S(x, y))$, so its negation is $\forall x \forall y\,(x \neq y \to \neg S(x, y))$. (The old key wrote the statement with $\to$ and gave (B) as the negation.)

My answer

Q66.

Quiz 1 Alkhulaifi

The domain is a set of animals on a farm, $H(x)$: "x is a horse", $F(x)$: "x has been fed". "Exactly one horse has not been fed" is:

(A) $\forall x \forall y\,\big((H(y) \wedge x \neq y) \to F(y)\big)$

(B) $\exists x \forall y\,\big(H(x) \wedge \neg F(x) \wedge F(y)\big)$

(C) $\exists x\,\big(H(x) \wedge \neg F(x)\big)$

(D) $\exists x \forall y\,\big(H(x) \wedge \neg F(x) \wedge ((H(y) \wedge y \neq x) \to F(y))\big)$

Answer: (D). Some horse x was not fed, and every other horse y was fed. (C) only says "at least one".

My answer

Q67.

[new]

The domain is a group of people, $M(x, y)$: "x sent an email to y". "Everyone sent an email to everyone else" is:

(A) $\forall x \forall y\, M(x, y)$

(B) $\forall x \forall y\,\big((x \neq y) \wedge M(x, y)\big)$

(C) $\forall x \forall y\,\big((x \neq y) \to M(x, y)\big)$

(D) $\forall x \exists y\,\big((x \neq y) \wedge M(x, y)\big)$

Answer: (C). The conditional excludes the pairs with $x = y$. (A) also requires emails to oneself; (B) is false whenever $x = y$; (D) means "someone else".

My answer

Q68.

Quiz 2 (232) Sec 3

Logical fallacies are errors in reasoning that make an argument invalid. Which of the following is a fallacy?

(I) $((p \to q) \wedge q) \to p$ (II) $((p \to q) \wedge \neg p) \to \neg q$ (III) $((p \to q) \wedge \neg q) \to \neg p$

(A) Only I and II

(B) Only I

(C) Only II

(D) Only II and III

Answer: (A). (I) affirms the conclusion and (II) denies the hypothesis: neither is a tautology (take $p = \text{F}$, $q = \text{T}$). (III) is modus tollens.

My answer

Q69.

Quiz 2 (232) Sec 8

The proposition $\big((p \to q) \wedge \neg p\big) \to \neg q$ is known as:

(A) hypothetical syllogism

(B) modus ponens

(C) modus tollens

(D) a fallacy

Answer: (D). It is the inverse error (denying the hypothesis): it is false for $p = \text{F}$, $q = \text{T}$.

My answer

Q70.

HW2 (261), Rosen 1.6 #2

Consider the argument: "If George does not have eight legs, then he is not a spider. George is a spider. $\therefore$ George has eight legs." Which statement is correct?

(A) Valid (modus tollens); if the premises are true, the conclusion must be true.

(B) Invalid (affirming the conclusion).

(C) Valid (modus ponens), but the conclusion may still be false when the premises are true.

(D) Invalid (denying the hypothesis).

Answer: (A). With p: "George has eight legs" and q: "George is a spider", the form is $\neg p \to \neg q,\ q\ \therefore p$, which is modus tollens.

My answer

Q71.

HW2 (261), Rosen 1.6 #19a

"If n is a real number such that $n > 1$, then $n^2 > 1$. Suppose that $n^2 > 1$. Then $n > 1$." This argument is

(A) valid, by modus ponens

(B) valid, by modus tollens

(C) invalid: the fallacy of affirming the conclusion

(D) invalid: the fallacy of denying the hypothesis

Answer: (C). The form is $p \to q,\ q\ \therefore p$; for example $n = -2$ has $n^2 > 1$ but $n < 1$.

My answer

Q72.

HW2 (261), Rosen 1.6 #19(b,c)

(b) "If $n > 3$, then $n^2 > 9$. Suppose $n^2 \leq 9$. Then $n \leq 3$." (c) "If $n > 2$, then $n^2 > 4$. Suppose $n \leq 2$. Then $n^2 \leq 4$." (n real.)

(A) (b) is valid by modus tollens; (c) is invalid (denying the hypothesis).

(B) (b) is invalid; (c) is valid by modus tollens.

(C) Both are valid.

(D) (b) is valid by modus ponens; (c) is invalid (affirming the conclusion).

Answer: (A). (b) is $p \to q,\ \neg q\ \therefore \neg p$. (c) is $p \to q,\ \neg p\ \therefore \neg q$; $n = -3$ is a counterexample.

My answer

Q73.

[new]

Which truth assignment proves that the argument $p \to q,\ q \to r,\ r\ \therefore p$ is invalid?

(A) $p = \text{T}, q = \text{T}, r = \text{T}$

(B) $p = \text{F}, q = \text{F}, r = \text{T}$

(C) $p = \text{F}, q = \text{F}, r = \text{F}$

(D) $p = \text{T}, q = \text{F}, r = \text{T}$

Answer: (B). All three hypotheses are true and the conclusion p is false. In (A) the conclusion is true; in (C) r is false; in (D) $p \to q$ is false.

My answer

Q74. Quiz 1 Alkhulaifi

"If I work all night on this homework, then I can answer all the exercises. If I answer all the exercises, I will understand the material. Therefore, if I work all night on this homework, then I will understand the material." Which statement is correct?

(A) The argument is valid by directly applying hypothetical syllogism.

(B) The argument is valid by directly applying modus tollens.

(C) The argument is valid by directly applying disjunctive syllogism.

(D) The argument is not valid.

Answer: (A). The form is $p \to q,\ q \to r\ \therefore p \to r$.

My answer

Q75. Quiz 2 (232) Sec 8

"Tom and Jerry are famous cartoon characters. Therefore, Jerry is a famous cartoon character." Which rule of inference is used?

(A) Addition

(B) Simplification

(C) Disjunctive syllogism

(D) Conjunction

Answer: (B). From $T \wedge J$ we conclude J.

My answer

Q76. HW2 (261), Rosen 1.6 #4a

"Kangaroos live in Australia and are marsupials. Therefore, kangaroos are marsupials." Which rule of inference is used?

(A) Simplification

(B) Modus tollens

(C) Addition

(D) Conjunction

Answer: (A). $p \wedge q\ \therefore q$. (A homework answer said modus tollens.)

My answer

Q77. HW2 (261), Rosen 1.6 #4(b,c,d)

Name the rule used in each argument. (b) "It is either hotter than 100 degrees today or the pollution is dangerous. It is less than 100 degrees outside today. Therefore, the pollution is dangerous." (c) "Linda

is an excellent swimmer. If Linda is an excellent swimmer, then she can work as a lifeguard. Therefore, Linda can work as a lifeguard." (d) "Steve will work at a computer company this summer. Therefore, this summer Steve will work at a computer company or he will be a beach bum."

(A) (b) disjunctive syllogism, (c) modus ponens, (d) addition

(B) (b) modus tollens, (c) modus ponens, (d) simplification

(C) (b) resolution, (c) modus ponens, (d) conjunction

(D) (b) disjunctive syllogism, (c) hypothetical syllogism, (d) addition

Answer: (A). (b) $p \vee q,\ \neg p\ \therefore q$; (c) $p,\ p \to q\ \therefore q$; (d) $p\ \therefore p \vee q$.

My answer

Q78.

HW2 (261), Rosen 1.6 #9b

Premises: "If I eat spicy foods, then I have strange dreams." "I have strange dreams if there is thunder while I sleep." "I did not have strange dreams." What relevant conclusion can be drawn?

(A) I did not eat spicy food and there was no thunder while I slept.

(B) I ate spicy food.

(C) There was thunder while I slept, but I did not eat spicy food.

(D) No conclusion can be drawn.

Answer: (A). $s \to d$ and $\neg d$ give $\neg s$, and $t \to d$ and $\neg d$ give $\neg t$ (modus tollens twice), then conjunction.

My answer

Q79.

HW2 (261), Rosen 1.6 #31

Use resolution to show that the hypotheses $\neg r \vee u$ ("It is not raining or Yvette has her umbrella"), $\neg u \vee \neg w$ ("Yvette does not have her umbrella or she does not get wet") and $r \vee \neg w$ ("It is raining or Yvette does not get wet") imply $\neg w$. Resolving the first two hypotheses gives which clause, so that resolving it with the third gives $\neg w \vee \neg w \equiv \neg w$?

(A) $\neg r \vee \neg w$

(B) $u \vee \neg w$

(C) $\neg r \vee w$

(D) $r \vee u$

Answer: (A). Resolve on u: $\neg r \vee u,\ \neg u \vee \neg w\ \therefore \neg r \vee \neg w$. Then resolve on r with $r \vee \neg w$ and use the idempotent law.

My answer

Q80.

[new]

"Maite is excited or Maite will not go to the movie. Maite will go to the movie. Therefore, Maite is excited." Which rule of inference (together with double negation) justifies this argument?

(A) Disjunctive syllogism

(B) Modus tollens

(C) Addition

(D) Hypothetical syllogism

Answer: (A). With e, m: $e \vee \neg m$ and $m \equiv \neg\neg m$ give e.

My answer

Q81.

Quiz 2 (232) Sec 8

"Norah is an engineer. Norah likes logic. Therefore, some engineers like logic." Which rules of inference are needed to prove this argument?

(A) Existential instantiation and modus ponens

(B) Conjunction and existential generalization

(C) Universal instantiation and modus ponens

(D) Simplification and universal generalization

Answer: (B). Conjunction gives $E(\text{Norah}) \wedge L(\text{Norah})$, and existential generalization gives $\exists x(E(x) \wedge L(x))$.

My answer

Q82.

Quiz 1 Alenzi

To establish the validity of $\exists x\, P(x),\ \forall x\, Q(x)\ \therefore \exists x\, (P(x) \wedge Q(x))$, the correct sequence of rules (after the hypotheses) is:

(A) Existential instantiation (particular c with $P(c)$), universal instantiation ($Q(c)$), conjunction, existential generalization

(B) Universal instantiation, universal generalization, conjunction, existential generalization

(C) Existential instantiation twice with the same c, conjunction, universal generalization

(D) Universal instantiation, existential instantiation, conjunction, universal generalization

Answer: (A). The particular c from $\exists x\, P(x)$ is defined first; $\forall x\, Q(x)$ can then be instantiated at that c. (In the key's proof some line references are off: Simplification is applied to line 3 and Conjunction to lines 5 and 6.)

My answer

Q83.

Quiz 2 Adel

"Every student who missed class or got a detention did not get an A. Saad is a student in the class. Saad got an A. $\therefore$ Saad did not get a detention." Form: $\forall x\, ((M(x) \vee D(x)) \rightarrow \neg A(x))$, $A(\text{Saad})\ \therefore \neg D(\text{Saad})$. After universal instantiation gives $(M(\text{Saad}) \vee D(\text{Saad})) \rightarrow \neg A(\text{Saad})$, which rules complete the proof?

(A) Double negation, modus tollens, De Morgan's law, simplification

(B) Modus ponens, addition

(C) Disjunctive syllogism, conjunction

(D) Hypothetical syllogism, simplification

Answer: (A). $A(\text{Saad}) \equiv \neg\neg A(\text{Saad})$; modus tollens gives $\neg(M \vee D)$; De Morgan gives $\neg M \wedge \neg D$; simplification (after commuting) gives $\neg D(\text{Saad})$.

My answer

Q84.

HW2 (261), Rosen 1.6 #9f

Premises: "All rodents gnaw their food." "Mice are rodents." "Rabbits do not gnaw their food." "Bats are not rodents." Which conclusions can validly be drawn?

(A) Mice gnaw their food, and rabbits are not rodents.

(B) Bats do not gnaw their food.

(C) Rabbits are rodents.

(D) Mice gnaw their food, rabbits are not rodents, and bats do not gnaw their food.

Answer: (A). Universal instantiation with modus ponens (mice) and with modus tollens (rabbits). Nothing follows about bats: concluding that they do not gnaw would deny the hypothesis.

My answer

Q85.

HW2 (261), Rosen 1.6 #13b

"Somebody in this class enjoys whale watching. Every person who enjoys whale watching cares about ocean pollution. Therefore, there is a person in this class who cares about ocean pollution." Which rules are used, in order?

(A) Existential instantiation, universal instantiation, modus ponens, existential generalization

(B) Universal instantiation, universal generalization, modus ponens, existential generalization

(C) Existential instantiation, universal instantiation, modus tollens, existential generalization

(D) Universal instantiation, existential instantiation, modus ponens, universal generalization

Answer: (A). Take a particular c with $W(c)$; $W(c) \to P(c)$ by universal instantiation; then $P(c)$ by modus ponens, and $\exists x\, P(x)$.

My answer

Q86.

HW2 (261), Rosen 1.6 #13d

"Everyone in New Jersey lives within 50 miles of the ocean. Someone in New Jersey has never seen the ocean. Therefore, someone who lives within 50 miles of the ocean has never seen the ocean." Which rules are used?

(A) Existential instantiation, simplification, universal instantiation, modus ponens, conjunction, existential generalization

(B) Existential instantiation, universal instantiation, modus tollens, conjunction, existential generalization

(C) Universal instantiation, universal generalization, modus ponens, conjunction

(D) Existential instantiation, simplification, universal instantiation, disjunctive syllogism, universal generalization

Answer: (A). From $J(c) \wedge \neg S(c)$ get $J(c)$; $J(c) \rightarrow W(c)$ gives $W(c)$; then $W(c) \wedge \neg S(c)$, and existential generalization.

My answer

Q87.

HW2 (261), Rosen 1.6 #15

(a) "All students in this class understand logic. Xavier is a student in this class. Therefore, Xavier understands logic." (b) "Every computer science major takes discrete mathematics. Natasha is taking discrete mathematics. Therefore, Natasha is a computer science major." (c) "All parrots like fruit. My pet bird is not a parrot. Therefore, my pet bird does not like fruit." Which arguments are correct?

(A) Only (a)

(B) (a) and (b) only

(C) Only (c)

(D) All three

Answer: (A). (a) universal instantiation and modus ponens. (b) affirms the conclusion and (c) denies the hypothesis.

My answer

Q88.

HW2 (232), Rosen 1.6 #16a

"Everyone enrolled in the university has lived in a dormitory. Mia has never lived in a dormitory. Therefore, Mia is not enrolled in the university." This argument is:

(A) Correct, by universal instantiation and modus tollens

(B) Correct, by universal instantiation and modus ponens

(C) Incorrect: the fallacy of denying the hypothesis

(D) Incorrect: the fallacy of affirming the conclusion

Answer: (A). $E(\text{Mia}) \rightarrow D(\text{Mia})$ and $\neg D(\text{Mia})$ give $\neg E(\text{Mia})$.

My answer

Q89. HW2 (232), Rosen 1.6 #16(b,c,d)

(b) "A convertible car is fun to drive. Isaac's car is not a convertible. Therefore, Isaac's car is not fun to drive." (c) "Quincy likes all action movies. Quincy likes the movie *Eight Men Out*. Therefore, *Eight Men Out* is an action movie." (d) "All lobstermen set at least a dozen traps. Hamilton is a lobsterman. Therefore, Hamilton sets at least a dozen traps."

(A) (b) incorrect, (c) incorrect, (d) correct

(B) (b) correct, (c) incorrect, (d) correct

(C) All three incorrect

(D) (b) incorrect, (c) correct, (d) correct

Answer: (A). (b) denies the hypothesis; (c) affirms the conclusion; (d) uses universal instantiation and modus ponens.

My answer

Q90. HW2 (261), Rosen 1.6 #23

The following "proof" supposedly shows that if $\exists x\, P(x) \wedge \exists x\, Q(x)$ is true then $\exists x\, (P(x) \wedge Q(x))$ is true.

1.	$\exists x\, P(x) \vee \exists x\, Q(x)$	Premise
2.	$\exists x\, P(x)$	Simplification from (1)
3.	$P(c)$	Existential instantiation from (2)
4.	$\exists x\, Q(x)$	Simplification from (1)
5.	$Q(c)$	Existential instantiation from (4)
6.	$P(c) \wedge Q(c)$	Conjunction from (3) and (5)
7.	$\exists x\, (P(x) \wedge Q(x))$	Existential generalization

The essential error is:

(A) Line 5: existential instantiation must introduce a new element, so it cannot reuse the c from line 3.

(B) Line 3: existential instantiation cannot be applied to $\exists x\, P(x)$.

(C) Line 6: conjunction cannot be applied to $P(c)$ and $Q(c)$.

(D) Line 7: it should be universal generalization.

Answer: (A). Each use of existential instantiation needs its own new name, e.g. $Q(d)$. (As written, lines 2 and 4 are also wrong: simplification cannot be applied to the disjunction in line 1.)

My answer

Part B: Written problems

Q91.

Quiz 1 Alrasasi

Construct the truth table for $(p \to q) \to (p \vee r \to q \vee r)$. What kind of proposition is it?

Answer:

By precedence, $p \vee r \to q \vee r$ means $(p \vee r) \to (q \vee r)$. Call the whole proposition $(*)$.

p	q	r	$p \to q$	$p \vee r$	$q \vee r$	$p \vee r \to q \vee r$	$(*)$
T	T	T	T	T	T	T	T
T	T	F	T	T	T	T	T
T	F	T	F	T	T	T	T
T	F	F	F	T	F	F	T
F	T	T	T	T	T	T	T
F	T	F	T	F	T	T	T
F	F	T	T	T	T	T	T
F	F	F	T	F	F	T	T

The last column is T in every row, so $(p \to q) \to (p \vee r \to q \vee r)$ is a **tautology**. (The only row where $p \vee r \to q \vee r$ is false is $p = \text{T}, q = r = \text{F}$, and there the hypothesis $p \to q$ is also false.)

My answer

Q92.

HW1 (261), Rosen 1.3 #10b; HW1 (232), #12b

Show that $\big[(p \to q) \wedge (q \to r)\big] \to (p \to r)$ is a tautology using a truth table. Then explain why this justifies the rule of hypothetical syllogism.

Answer:

Let $H = (p \to q) \wedge (q \to r)$.

p	q	r	$p \to q$	$q \to r$	H	$p \to r$	$H \to (p \to r)$
T	T	T	T	T	T	T	T
T	T	F	T	F	F	F	T
T	F	T	F	T	F	T	T
T	F	F	F	T	F	F	T
F	T	T	T	T	T	T	T
F	T	F	T	F	F	T	T
F	F	T	T	T	T	T	T
F	F	F	T	T	T	T	T

Every entry of the last column is T, so the proposition is a tautology. In every row where both hypotheses $p \to q$ and $q \to r$ are true (rows 1, 5, 7, 8), the conclusion $p \to r$ is also true. That is exactly what it means for the argument $p \to q,\ q \to r\ \therefore p \to r$ (hypothetical syllogism) to be valid. ■

My answer

Q93. HW1 (261), Rosen 1.3 #33

Show that $(p \to q) \to (r \to s)$ and $(p \to r) \to (q \to s)$ are not logically equivalent.

Answer:

To show non-equivalence it is enough to give one truth assignment under which the two propositions have different values. Take

$$p = \text{F}, \quad q = \text{F}, \quad r = \text{T}, \quad s = \text{F}.$$

First proposition: $p \to q = \text{F} \to \text{F} = \text{T}$ and $r \to s = \text{T} \to \text{F} = \text{F}$, so $(p \to q) \to (r \to s) = \text{T} \to \text{F} = \text{F}$.

Second proposition: $p \to r = \text{F} \to \text{T} = \text{T}$ and $q \to s = \text{F} \to \text{F} = \text{T}$, so $(p \to r) \to (q \to s) = \text{T} \to \text{T} = \text{T}$.

The values differ, so the propositions are not logically equivalent. (The assignment $p = \text{F}, q = \text{T}, r = \text{F}, s = \text{F}$ also works.) There is no need for the full 16-row table. ■

My answer

Q94. Quiz 1 Alrasasi

Use the laws of propositional logic to prove $\neg p \to (q \to r) \equiv q \to (p \vee r)$.

Answer:

$$
\begin{aligned}
\neg p \to (q \to r) &\equiv \neg p \to (\neg q \vee r) && \text{Conditional identity} \\
&\equiv \neg\neg p \vee (\neg q \vee r) && \text{Conditional identity} \\
&\equiv p \vee (\neg q \vee r) && \text{Double negation law} \\
&\equiv (\neg q \vee r) \vee p && \text{Commutative law} \\
&\equiv \neg q \vee (r \vee p) && \text{Associative law} \\
&\equiv \neg q \vee (p \vee r) && \text{Commutative law} \\
&\equiv q \to (p \vee r) && \text{Conditional identity}
\end{aligned}
$$

■

My answer

Q95. Quiz 1 Adel

Write the negation of $\exists x \forall y \left(P(x, y) \leftrightarrow P(y, x)\right)$ so that all negations immediately precede predicates. The final answer may contain the conjunction and disjunction operators only (no $\to$ or $\leftrightarrow$).

Answer:

$$
\begin{aligned}
&\neg\exists x\forall y\,\big(P(x,y)\leftrightarrow P(y,x)\big)\\
\equiv\ &\forall x\,\neg\forall y\,\big(P(x,y)\leftrightarrow P(y,x)\big) && \text{De Morgan (quantifiers)}\\
\equiv\ &\forall x\exists y\,\neg\big(P(x,y)\leftrightarrow P(y,x)\big) && \text{De Morgan (quantifiers)}\\
\equiv\ &\forall x\exists y\,\neg\big[(P(x,y)\to P(y,x))\wedge(P(y,x)\to P(x,y))\big] && \text{Conditional identity}\\
\equiv\ &\forall x\exists y\,\big[\neg(P(x,y)\to P(y,x))\vee\neg(P(y,x)\to P(x,y))\big] && \text{De Morgan}\\
\equiv\ &\forall x\exists y\,\big[\neg(\neg P(x,y)\vee P(y,x))\vee\neg(\neg P(y,x)\vee P(x,y))\big] && \text{Conditional identity}\\
\equiv\ &\forall x\exists y\,\big[(P(x,y)\wedge\neg P(y,x))\vee(\neg P(x,y)\wedge P(y,x))\big] && \text{De Morgan, double negation}
\end{aligned}
$$

Final form: $\forall x\exists y\,\big[(P(x,y)\wedge\neg P(y,x))\vee(\neg P(x,y)\wedge P(y,x))\big]$, i.e. for every x there is a y such that exactly one of $P(x,y)$, $P(y,x)$ holds. ■

My answer

Q96.

Quiz 1 Alkhulaifi

Prove that the argument is valid by replacing each proposition with a variable to obtain the form of the argument. Then use the rules of inference to prove that the form is valid.

If it was not cloudy or it didn't rain (or both), then the picnic was held and photos were taken.
Photos were not taken.
$\therefore$ It rained.

Answer:

Variables: c: it was cloudy, r: it rained, p: the picnic was held, q: photos were taken. The form is

$$(\neg c\vee\neg r)\to(p\wedge q),\qquad \neg q\qquad \therefore\ r.$$

1.	$\neg q$	Hypothesis
2.	$\neg q\vee\neg p$	Addition, 1
3.	$\neg p\vee\neg q$	Commutative law, 2
4.	$\neg(p\wedge q)$	De Morgan's law, 3
5.	$(\neg c\vee\neg r)\to(p\wedge q)$	Hypothesis
6.	$\neg(\neg c\vee\neg r)$	Modus tollens, 4, 5
7.	$\neg\neg c\wedge\neg\neg r$	De Morgan's law, 6
8.	$c\wedge r$	Double negation law, 7
9.	$r\wedge c$	Commutative law, 8
10.	r	Simplification, 9

The last line is the conclusion, so the argument is valid. ■

My answer

Q97. HW2 (261), Rosen 1.6 #12

Show that the argument form with premises $(p \wedge t) \rightarrow (r \vee s)$, $q \rightarrow (u \wedge t)$, $u \rightarrow p$ and $\neg s$ and conclusion $q \rightarrow r$ is valid, using rules of inference and laws of propositional logic.

Answer:

1.	$q \rightarrow (u \wedge t)$	Hypothesis
2.	$\neg q \vee (u \wedge t)$	Conditional identity, 1
3.	$(\neg q \vee u) \wedge (\neg q \vee t)$	Distributive law, 2
4.	$\neg q \vee u$	Simplification, 3
5.	$q \rightarrow u$	Conditional identity, 4
6.	$u \rightarrow p$	Hypothesis
7.	$q \rightarrow p$	Hypothetical syllogism, 5, 6
8.	$(\neg q \vee t) \wedge (\neg q \vee u)$	Commutative law, 3
9.	$\neg q \vee t$	Simplification, 8
10.	$\neg q \vee p$	Conditional identity, 7
11.	$(\neg q \vee p) \wedge (\neg q \vee t)$	Conjunction, 10, 9
12.	$\neg q \vee (p \wedge t)$	Distributive law, 11
13.	$q \rightarrow (p \wedge t)$	Conditional identity, 12
14.	$(p \wedge t) \rightarrow (r \vee s)$	Hypothesis
15.	$q \rightarrow (r \vee s)$	Hypothetical syllogism, 13, 14
16.	$\neg q \vee (r \vee s)$	Conditional identity, 15
17.	$s \vee (\neg q \vee r)$	Associative and commutative laws, 16
18.	$\neg s$	Hypothesis
19.	$\neg q \vee r$	Disjunctive syllogism, 17, 18
20.	$q \rightarrow r$	Conditional identity, 19

The conclusion $q \rightarrow r$ is reached, so the argument form is valid. ■

My answer

Q98. Quiz 2 (232) Sec 3

Given the premises $\exists x\,(D(x) \wedge \neg E(x))$ and $\forall x\,(D(x) \rightarrow Q(x))$, show that we can reach the conclusion $\exists x\,(Q(x) \wedge \neg E(x))$, using only the premises and the rules of inference (simplification, modus ponens, conjunction, existential and universal instantiation and generalization).

Answer:

1.	$\exists x\,(D(x) \wedge \neg E(x))$	Hypothesis
2.	$D(a) \wedge \neg E(a)$, a a particular element	Existential instantiation, 1
3.	$D(a)$	Simplification, 2
4.	$\forall x\,(D(x) \rightarrow Q(x))$	Hypothesis
5.	$D(a) \rightarrow Q(a)$	Universal instantiation, 4 (element a)
6.	$Q(a)$	Modus ponens, 3, 5
7.	$\neg E(a) \wedge D(a)$	Commutative law, 2
8.	$\neg E(a)$	Simplification, 7
9.	$Q(a) \wedge \neg E(a)$	Conjunction, 6, 8
10.	$\exists x\,(Q(x) \wedge \neg E(x))$	Existential generalization, 9

Existential instantiation is used first, so that universal instantiation can then be applied to the same particular element a. ■

My answer

Q99. HW2 (261), Rosen 1.6 #28

Use rules of inference to show that if $\forall x\,(P(x) \vee Q(x))$ and $\forall x\,\big((\neg P(x) \wedge Q(x)) \rightarrow R(x)\big)$ are true, then $\forall x\,(\neg R(x) \rightarrow P(x))$ is also true (all domains the same).

Answer:

1.	c is an arbitrary element of the domain	Element definition
2.	$\forall x\,(P(x) \vee Q(x))$	Hypothesis
3.	$P(c) \vee Q(c)$	Universal instantiation, 1, 2
4.	$Q(c) \vee P(c)$	Commutative law, 3
5.	$\forall x\,((\neg P(x) \wedge Q(x)) \rightarrow R(x))$	Hypothesis
6.	$(\neg P(c) \wedge Q(c)) \rightarrow R(c)$	Universal instantiation, 1, 5
7.	$\neg(\neg P(c) \wedge Q(c)) \vee R(c)$	Conditional identity, 6
8.	$(P(c) \vee \neg Q(c)) \vee R(c)$	De Morgan's law, double negation, 7
9.	$\neg Q(c) \vee (P(c) \vee R(c))$	Commutative and associative laws, 8
10.	$P(c) \vee (P(c) \vee R(c))$	Resolution, 4, 9
11.	$P(c) \vee R(c)$	Associative and idempotent laws, 10
12.	$\neg\neg R(c) \vee P(c)$	Commutative and double negation laws, 11
13.	$\neg R(c) \rightarrow P(c)$	Conditional identity, 12
14.	$\forall x\,(\neg R(x) \rightarrow P(x))$	Universal generalization, 1, 13

Since c is arbitrary, universal generalization in line 14 is allowed. ■

My answer

Q100. HW2 (261), Rosen 1.6 #29

Use rules of inference to show that if $\forall x\,(P(x) \lor Q(x))$, $\forall x\,(\neg Q(x) \lor S(x))$, $\forall x\,(R(x) \to \neg S(x))$ and $\exists x\,\neg P(x)$ are true, then $\exists x\,\neg R(x)$ is true.

Answer:

1.	$\exists x\,\neg P(x)$	Hypothesis
2.	$\neg P(c)$, c a particular element	Existential instantiation, 1
3.	$\forall x\,(P(x) \lor Q(x))$	Hypothesis
4.	$P(c) \lor Q(c)$	Universal instantiation, 3 (element c)
5.	$Q(c)$	Disjunctive syllogism, 4, 2
6.	$\forall x\,(\neg Q(x) \lor S(x))$	Hypothesis
7.	$\neg Q(c) \lor S(c)$	Universal instantiation, 6
8.	$\neg\neg Q(c)$	Double negation law, 5
9.	$S(c)$	Disjunctive syllogism, 7, 8
10.	$\forall x\,(R(x) \to \neg S(x))$	Hypothesis
11.	$R(c) \to \neg S(c)$	Universal instantiation, 10
12.	$\neg\neg S(c)$	Double negation law, 9
13.	$\neg R(c)$	Modus tollens, 11, 12
14.	$\exists x\,\neg R(x)$	Existential generalization, 13

■

My answer